iLearn Practice Exam

10BR · Chapter / Topic 9

Cambridge IGCSE Computer Science 0478

Paper 2 Algorithms, Programming and Logic

Chapter 9: Databases

Name: _____

Class: _____

Date: _____

Time allowed: 1 hour 30 minutes

TOTAL MARKS: 75

Marks obtained: ..... / 75

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 75.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 9 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 A shop that sells books has set up a new database table called `BookList` to store book details. Part of this table is given.

CatNo	Title	Fiction	Author	PaperBack	Price	StockLevel
BK01	The Princes' Story	Yes	B Penn	Yes	4.50	500
BK02	The Princesses' Story	Yes	B Penn	Yes	4.50	350
BK03	Computer Science	No	Way Yu	Yes	19.99	20
BK04	The Modern World	No	P Patel	No	25.00	5
BK05	The Ancient World	Yes	P Patel	No	25.00	5
BK06	Computer Science	No	R Dale	Yes	27.35	8
BK07	The Princes' Story	Yes	B Penn	No	12.50	3
BK08	The Princesses' Story	Yes	B Penn	No	12.50	0
BK12	Famous Five	Yes	E Bly	Yes	2.75	45
BK15	Secret Seven	Yes	E Bly	Yes	2.75	25
BK16	The Last Knight	Yes	P Mann	Yes	5.99	7
BK17	The Dark Tower	Yes	P Mann	Yes	5.99	5
BK19	The Final Chase	Yes	P Mann	Yes	5.99	5
BK21	Maths Today Part 1	No	B Ward	Yes	6.75	25
BK22	Maths Today Part 2	No	B Ward	Yes	6.75	15
BK23	Maths Today Part 3	No	B Ward	Yes	6.75	10
BK26	Maths Today Workbook	No	B Ward	Yes	6.75	30
BK27	Knitting for Beginners	No	A Smith	Yes	6.99	3
BK30	Woodwork for Beginners	No	A Smith	Yes	6.99	4
BK31	Networking for Beginners	No	A Smith	Yes	6.99	0

(a) State the number of records in this part of the database table.

..... [1]

(b) **(i)** Give the name of the field that would be used for the primary key.

..... [1]

(ii) State the reason for choosing this field for the primary key.

.....

..... [1]

(c) Complete the table to identify the most appropriate data type for each field based on the data shown in the table `BookList`

Field	Data type
`CatNo`	
`Title`	
`Fiction`	
`Price`	

[2]

(d) Write the output from this structured query language (SQL) statement.

```
SELECT CatNo, Title, Author
FROM BookList
WHERE StockLevel = 0;
```

.....

.....

.....

..... [2]

(e) Complete this SQL statement to display all the titles by the author B Penn.

```
SELECT .....
FROM .....
WHERE .....;
```

[2]

2 A database table called `TVRange` shows the main features and prices of a range of televisions.

TVCode	ScreenSize	Satellite	SmartTV	SoundBar	Price$
TV90SaSmSd	90	YES	YES	YES	9750.00
TV75SaSmSd	75	YES	YES	YES	8500.00
TV75SaSd	75	YES	NO	YES	8000.00
TV65SaSmSd	65	YES	YES	YES	6000.00
TV65SmSd	65	NO	YES	YES	5000.00
TV65SaSd	65	YES	NO	YES	5000.00
TV55SaSmSd	55	YES	YES	YES	4000.00
TV55SaSd	55	YES	NO	YES	3500.00
TV55SmSd	55	NO	YES	YES	3500.00
TV50SaSmSd	50	YES	YES	YES	2500.00
TV50Sa	50	YES	NO	NO	1750.00
TV50Sm	50	NO	YES	NO	1750.00
TV40Sa	40	YES	NO	NO	1200.00
TV40	40	NO	NO	NO	950.00
TV32	32	NO	NO	NO	650.00

(a) Give the name of the field that is most suitable to be the primary key.

State the reason for this choice.

Field .....

Reason .....

.....

[2]

(b) The database uses the data types:

- text
- character
- Boolean
- integer
- real
- date/time.

Complete the table to show the most appropriate data type for each field.
Each data type must be different.

Field	Data type
TVCode	
ScreenSize	
SmartTV	
Price$	

[2]

(c) Complete the structured query language (SQL) query to return the television (TV) code, screen size and price of all Smart TVs in the database table.

```
SELECT TVCode, ....., .....

..... TVRange

WHERE SmartTV = .....;
```

[4]

3 A model shop wants to set up a database to help with stock control of the model figures available for sale. The shop wants to store this information about the model figures:
Field 1 – catalogue number, for example MD1234
Field 2 – description, for example 'small white dog'
Field 3 – number in stock, for example 5
Field 4 – the price of each model, for example 7.40
Field 5 – if the model has already been painted, yes or no.

(a) The shop needs **five** fields for each record.
Give a suitable name and data type for each field.

Field 1 name .....

Data type .....

Field 2 name .....

Data type .....

Field 3 name .....

Data type .....

Field 4 name .....

Data type .....

Field 5 name .....

Data type .....

[5]

(b) **(i)** Give the name of the field that should be used for the primary key.

..... [1]

(ii) State why this field is used as the primary key.

..... [1]

(c) Structured query language (SQL) is used to query data stored in this database.
State what these SQL commands are used for.

`SELECT` .....

.....

`FROM` .....

.....

`WHERE` .....

.....

[3]

4 A database table called `Hangar1` stores details of a collection of historic aircraft at a museum.

ID	Make	Model	Year	Engines	Airworthy
JM1	Hawker Siddeley	Nimrod	1966	4	Y
JM2	Douglas	DC-10	1970	3	Y
JM3	Aérospatiale-BAC	Concorde	1973	4	N
PB1	De Havilland	DH-9	1918	1	Y
PB2	Hawker	Fury	1935	1	Y
PB3	Hawker	Nimrod	1934	1	Y
PM1	Fieseler	Storch	1942	1	N
PM2	Hawker	Hurricane	1942	1	Y
PM3	Supermarine	Spitfire	1942	1	N
PM4	Douglas	C-47 Dakota	1942	2	N
PS1	Boeing	314 Clipper	1936	4	N

(a) State the number of fields and records in the database table.

Fields .....

Records .....

[2]

(b) Give the output that would be produced by the structured query language (SQL) statement:

```
SELECT Model, Year, Engines
FROM Hangar1
WHERE Engines > 1
ORDER BY Year;
```

.....

.....

.....

.....

.....

.....

.....

..... [4]

(c) Complete the SQL statement to display only the ID and model of all the aircraft that are airworthy.

```
SELECT ID, .....
.....
.....
..... ;
```

[4]

MASTER · iLEARN CS · LEARN · PRACTISE · iLearn CS

5 Complete the paragraph about databases. Use terms from the list. You may need to use a term more than once. Some of the terms in the list will **not** be used.

fields	**comments**	**columns**	**constant**	**key**
table	**primary key**	**program**	**records**	**rows**
scripts	**database**	**validation**	**variable**	

Database tables consist of ..... and ..... .

Rows are ..... .

..... are fields.

Structured query language (SQL) ..... are used to query data.

A ..... uniquely identifies a record.

[6]

MASTER · iLEARN CS · LEARN · PRACTISE · iLearn CS

6 A bus company has set up a new database table called `BUS` to store details of the buses leaving the main bus terminal.

RouteID	Destination	Stops	Driver	Time	Return
BC01	Town Centre	10	B Smith	9:00	Yes
BC02	The Fountain	5	A Patel	9:10	Yes
BC03	The Palace	9	B Hayes	9:28	No
BC04	City Park	12	W Gao	9:55	Yes
BC05	City Mall	9	R Hussein	10:10	No
BC06	Swan Lake	14	S Clarke	10:33	No
BC07	Guildhall Square	7	J Page	11:34	No
BC08	Sunset Quay	6	A Patel	12:01	Yes
BC10	Fort Worthington	15	E Carr	12:01	Yes
BC12	The Observatory	18	B Smith	13:27	No
BC14	East Seaport	11	B Scott	14:07	No
BC15	The Football Ground	12	J Page	14:46	Yes
BC19	The Cricket Ground	11	B Hayes	15:14	No
BC23	Temple Ruins	14	W Gao	16:16	No
BC27	The University	9	A Patel	16:45	No

(a) (i) Identify **one** appropriate data type for the field `Return`

..... [1]

(ii) State the reason for the data type chosen.

.....

..... [1]

(b) Write the output from the structured query language (SQL) statement:

```
SELECT RouteID, Destination, Time

FROM BUS

WHERE Driver = "A Patel";
```

.....

.....

.....

.....

[2]

(c) Complete the SQL statement to display only the destination of the buses that have a return.

```
SELECT .....
.....
.....
..... [3]
```

7 **Four** descriptions of structured query language (SQL) keywords and **five** SQL keywords are given.

Draw **one** line from each description to its matching keyword.
Not all SQL keywords will be used.

Description	SQL keywords
identifies the database table	ORDER BY
sets the condition	COUNT
sorts the results	WHERE
returns the number of records	SUM
	FROM

[4]

8 A database table `WATERFOWL` stores some details of a range of birds found in the UK that live on or near water. The length and wingspan dimensions are in centimetres and the typical lifespan is in years.

Species	Family	Length	Wingspan	CoastalHabitat	Lifespan
Barnacle goose	Goose	63.5	130.0	FALSE	14
Bewick's swan	Swan	120.2	190.5	TRUE	8
Brent goose	Goose	58.0	94.3	TRUE	11
Canada goose	Goose	78.5	160.0	FALSE	6
Eider	Duck	58.0	94.5	TRUE	8
Goldeneye	Duck	44.5	72.1	TRUE	6
Goosander	Duck	63.3	90.0	TRUE	7
Long-tailed duck	Duck	44.0	76.6	TRUE	5
Mallard	Duck	59.7	90.8	FALSE	3
Mute swan	Swan	150.0	220.2	FALSE	10
Pink-footed goose	Goose	69.6	150.0	TRUE	8
Pintail	Duck	60.0	88.6	TRUE	3
Smew	Duck	41.5	62.1	FALSE	7
White-fronted goose	Goose	71.5	140.0	FALSE	6
Whooper swan	Swan	150.0	230.0	TRUE	9

(a) State the purpose of a primary key in a database table.

.....

..... [1]

(b) Complete the table for each of the given data types, to identify **one** field from the table `WATERFOWL` that is most likely to use it.

Each field must be different.

Data type	Field
Boolean	
Integer	
Real	
Text	

[2]

(c) Give the output that would be produced by the structured query language (SQL) statement:

```
SELECT Species, Length, Wingspan
FROM WATERFOWL
WHERE Family = "Swan"
ORDER BY Lifespan;
```

.....

.....

.....

.....

.....

..... [3]

(d) Complete the SQL statement to list only the species, family and lifespan of all waterfowl whose typical lifespan is at least 10 years, sorted in order of species.

SELECT .....

.....

.....

..... [4]

9 A database table `OPTION_SUBJECTS` stores some data about a number of subjects taken by students in a college.

field	description
`SubjectCode`	a code such as CSIG01 to represent the subject and level
`SubjectName`	the name of the subject, for example computer science
`Level`	the level of the course, for example IGCSE or A Level
`CourseWork`	whether or not coursework is included as part of the assessment
`PracticalExams`	the number of practical exams taken during the course
`TheoryExams`	the number of theory exams taken during the course
`CourseLength`	length of the course, for example 1 or 2, representing number of years
`MinimumClassSize`	the minimum number of students required for the course to run, for example 10
`MaximumClassSize`	the maximum number of students admitted to each class, for example 30

(a) Identify the most appropriate field in `OPTION_SUBJECTS` for a primary key.
Give **one** reason for your choice.

primary key field .....

reason .....

.....

[2]

(b) Identify **one** appropriate field from `OPTION_SUBJECTS` for each of the given data types.
State **one** reason for your choice in each case.

integer data type .....

reason .....

.....

Boolean data type .....

reason .....

.....

[4]

(c) Complete the structured query language (SQL) statement to list only the subject code, subject name and level for the minimum class size of less than 10. The results should be listed in order of subject code, from the lowest to the highest.

```
SELECT .....

FROM .....

WHERE .....

..... ;
```

[5]

END OF QUESTION PAPER

MASTER · iLEARN CS · LEARN · PRACTISE · iLearn CS